Higher-Order Fourier Neural Operator: Additional Experiments and Analysis

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1. Ablation on training, inference Wall clock time and peak memory usage.

Figure 1. **Ablation on training time, inference time and peak memory usage.** We report results for FNO with improved backbone, HO-FNO, LaMO and Transolver. The reported quatities are measured on a GPU Nvidia A100 with the configurations used in the experiments in Table 1. The results show that HO-FNO is consistently faster than LaMO at inference time, while also being faster to train and more memory-efficient on two out of three datasets. Compared to FNO, HO-FNO incurs no additional computational cost. In contrast, Transolver is the least efficient across all metrics.

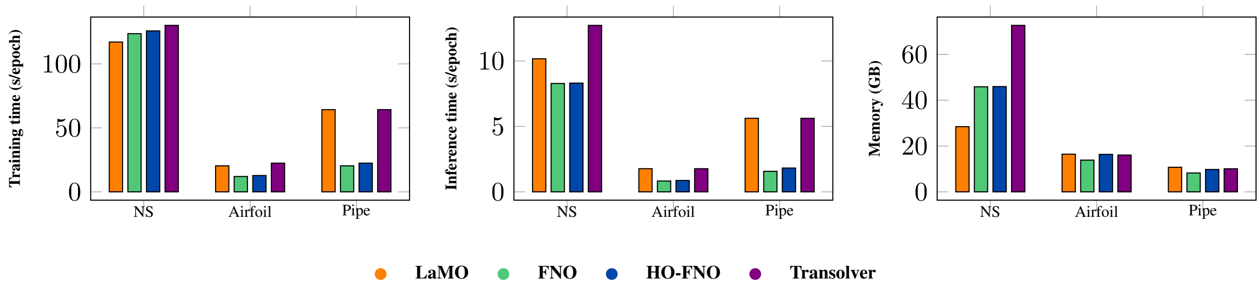
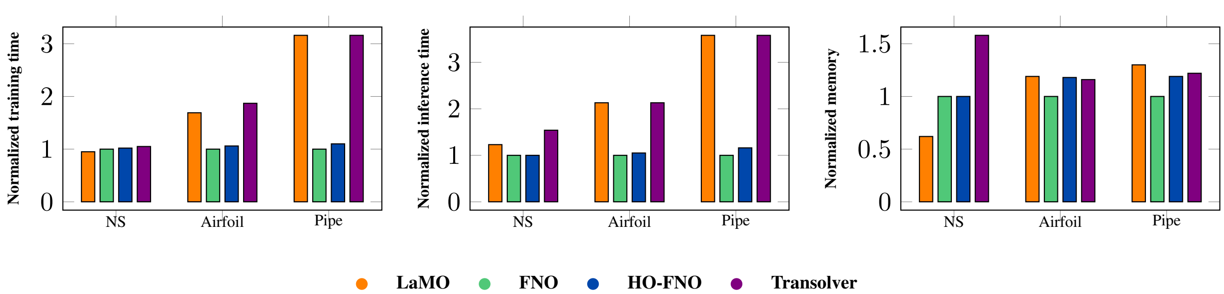


Figure 2. **Ablation on Normalized training time, inference time and peak memory usage.** We report results for FNO with improved backbone, HO-FNO, LaMO and Transolver. The reported quatities are measured on a GPU Nvidia A100 with the configurations used in the experiments in Table 1. In contrast to the previous visualization, each quantity is normalized with respect to FNO to facilitate comparison.



2. Spectral Analysis

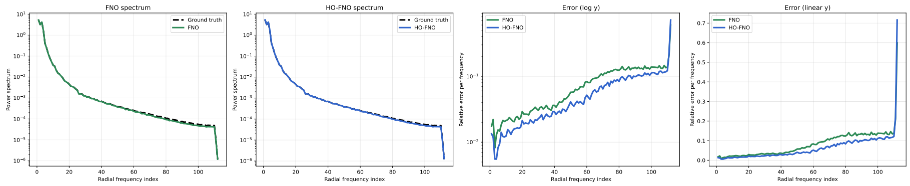


Figure 3. **Spectral Analysis on the Airfoil.** The first two plots show ground-truth and predicted energy spectra for FNO with improved backbone and HO-FNO, while the last two report frequency-wise errors. HO-FNO provides a closer match to the true spectrum and consistently lower errors across frequency bands.

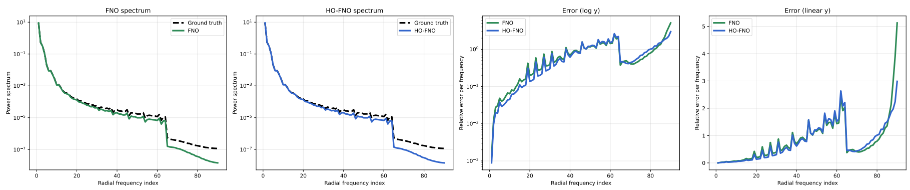


Figure 4. **Spectral Analysis on the Pipe.** The first two plots show ground-truth and predicted energy spectra for FNO with improved backbone and HO-FNO, while the last two report frequency-wise errors. HO-FNO provides a closer match to the true spectrum and reduces error across frequencies, with notable improvements at low and mid frequencies..

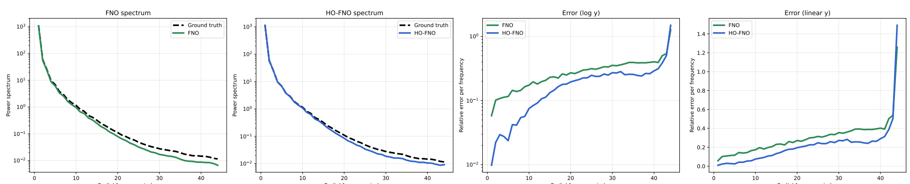


Figure 5. **Spectral Analysis on Navier–Stokes.** The first two plots show ground-truth and predicted energy spectra for FNO with improved backbone and HO-FNO, while the last two report frequency-wise errors. HO-FNO provides a closer match to the true spectrum and consistently lower errors across frequency bands.

3. Qualitative analysis throught visualizations of the predictions.

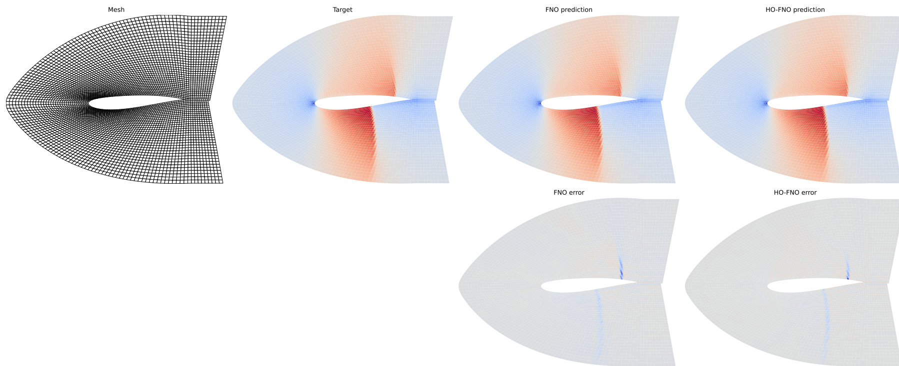


Figure 6. **Qualitative comparison on the Airfoil dataset.** From left to right: computational mesh, ground truth, FNO prediction, and HO-FNO prediction. Bottom: corresponding error maps. HO-FNO better captures flow structures around the airfoil and yields reduced errors compared to FNO with improved backbone.

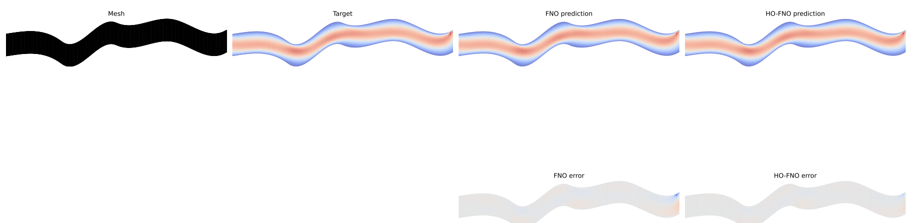


Figure 7. **Qualitative comparison on the Pipe dataset.** From left to right: computational mesh, ground truth, FNO prediction, and HO-FNO prediction. Bottom: corresponding error maps. HO-FNO more accurately captures the flow along the curved geometry and reduces errors compared to FNO with improved backbone.

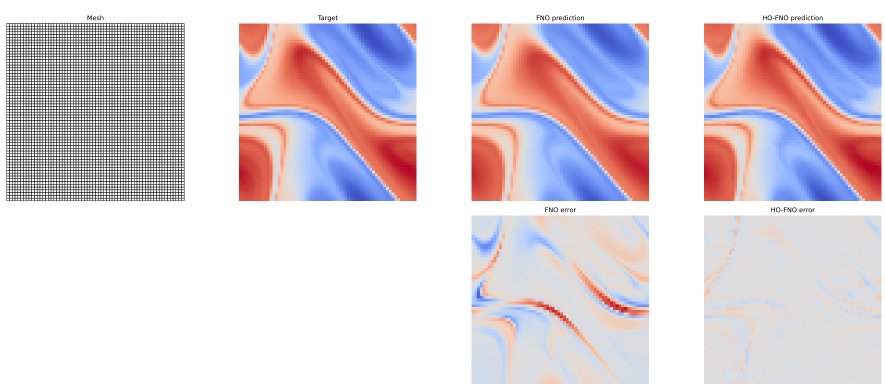


Figure 8. **Qualitative comparison on the Navier–Stokes dataset.** From left to right: input/mesh, ground truth, FNO prediction, and HO-FNO prediction. Bottom: corresponding error maps. HO-FNO better captures fine-scale flow structures and reduces errors compared to FNO with improved backbone.

4. Comparison with DPOT

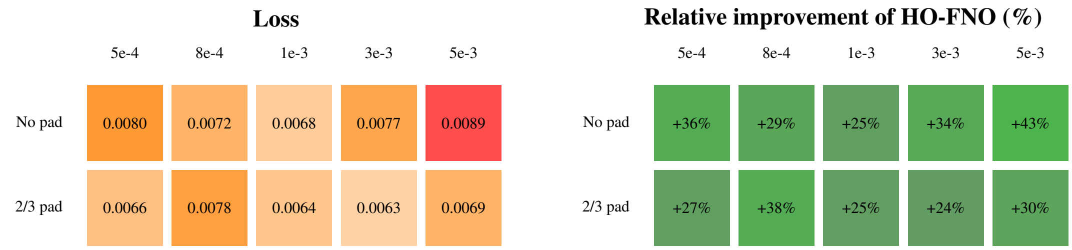


Figure 9. **Ablation on comparison with DPOT on the Airfoil dataset.** We report the test loss (left) and the relative improvement of HO-FNO over DPOT (right) across different learning rates. While $2/3$ padding consistently yields lower losses, HO-FNO provides substantial improvements over DPOT in both settings, with gains ranging from $+24\%$ to $+43\%$, demonstrating a wide improvement of HO-FNO over DPOT under an extensive tuning. DPOT was trained with the same parameter count as HO-FNO, using the hyperparameters specified in the Appendix of the submitted paper.